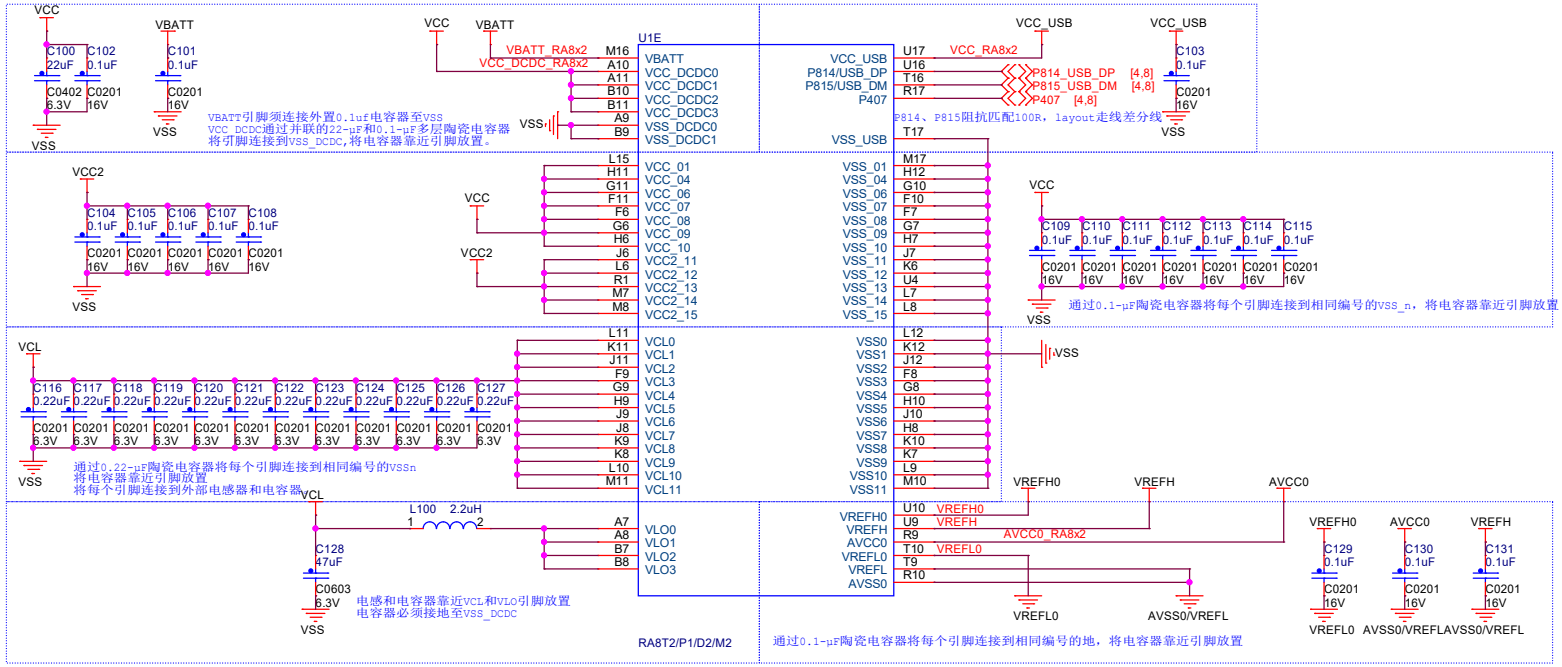
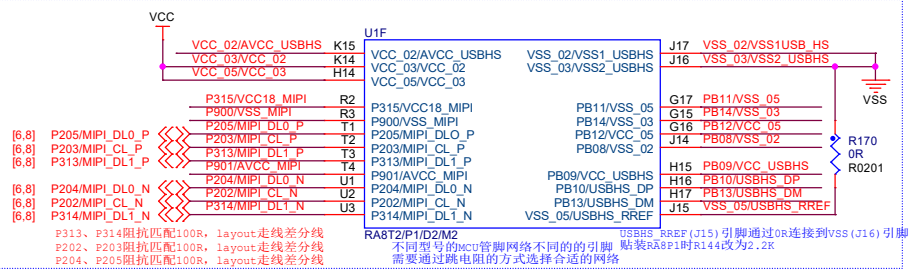
Page1: MCU 电源
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Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB



核心板缺省功能阻容设置

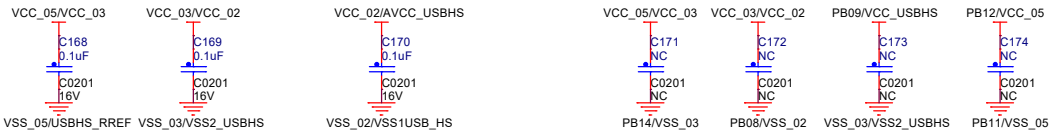
核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	0.1uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1UF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R@100M
R172-R180	NC	0R
R191-R193	NC	0R



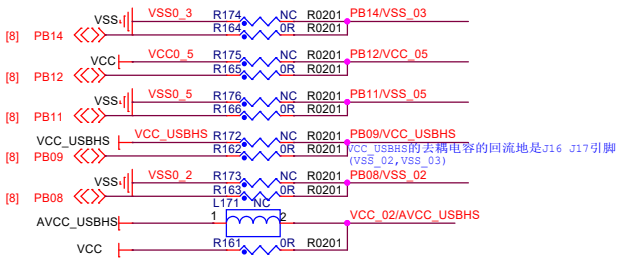
由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8T2

电阻选装

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放

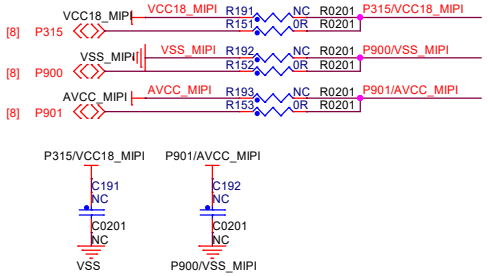


与电源相关的引脚，MCU信号出来后立即用电阻分离



MIPI部分

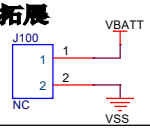
要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放



差分线走到J1(BTB)附近再用电阻分离。

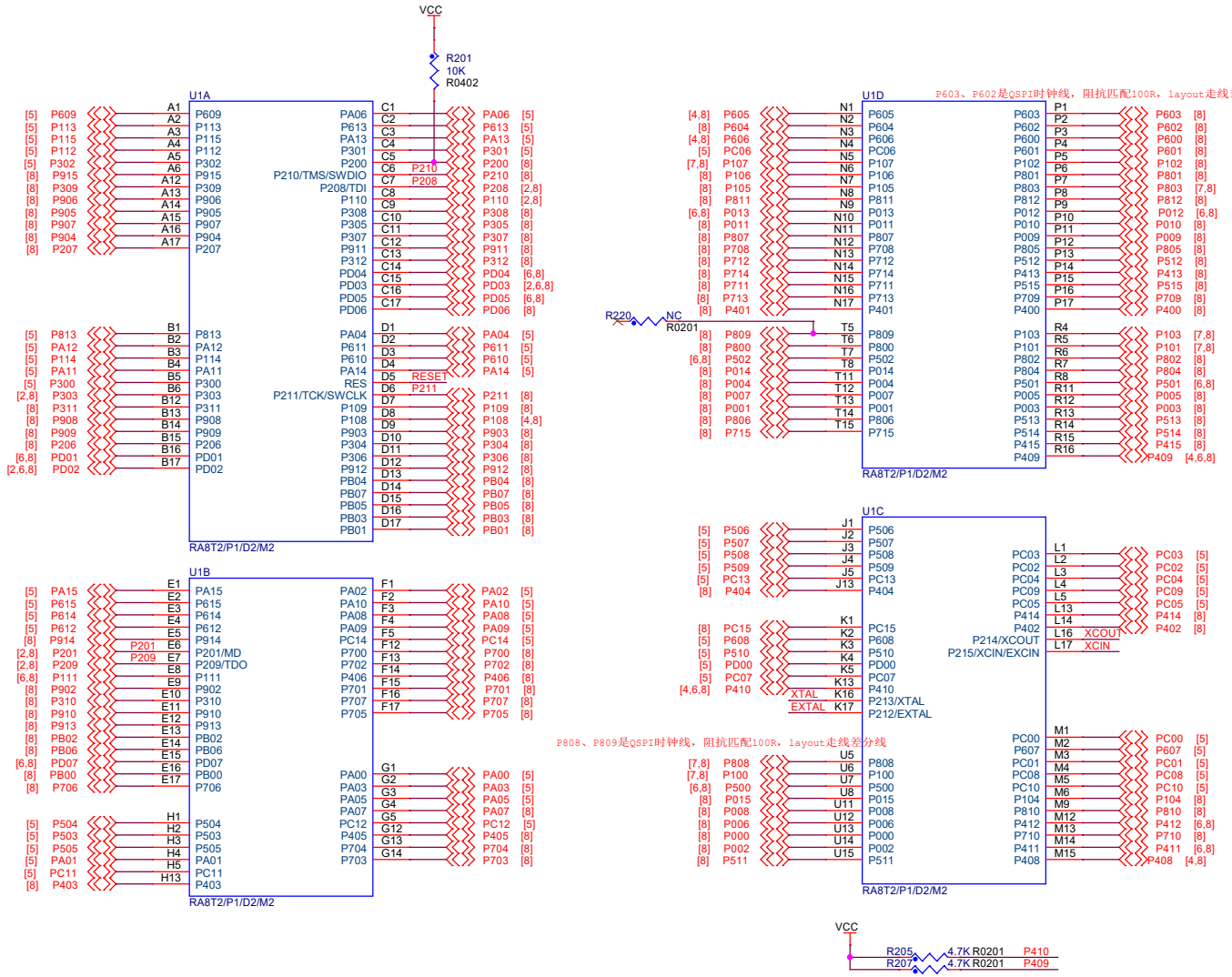


通孔拓展

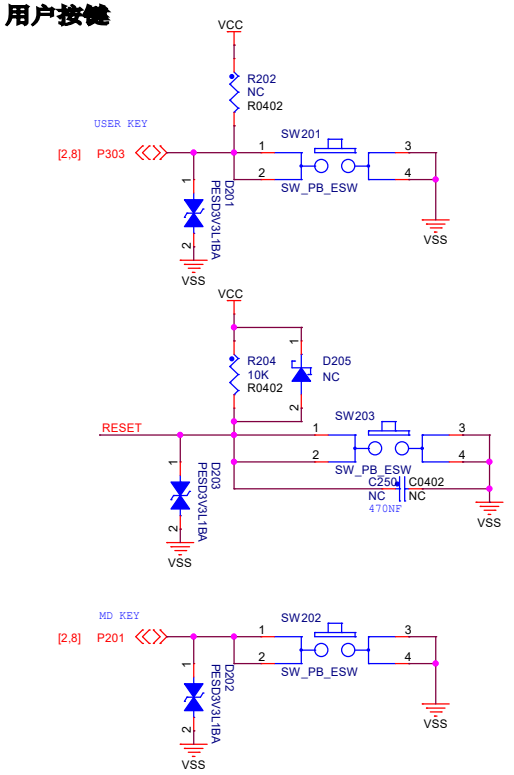


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Title	野火瑞萨CPKNET-RA8T2L 核心板		
Size A3	Document Number MCU 电源		Rev V1.2
Date: Monday, June 08, 2026		Sheet 1 of 10	

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Page5: SDRAM
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Page7: 电源LDO/DCDC/MIPI/FLASH
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用户按键

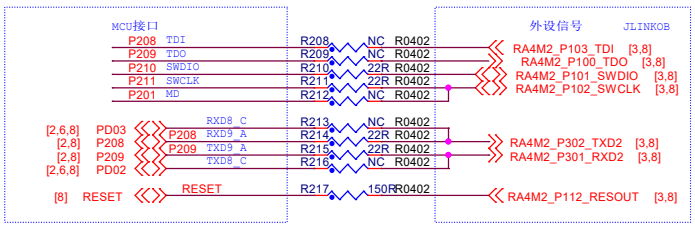


LED

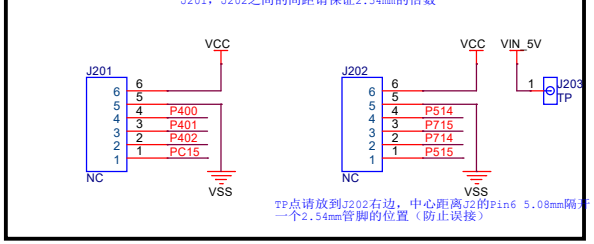


如果使用USB HS，主晶振只能选12/20/24/48MHz

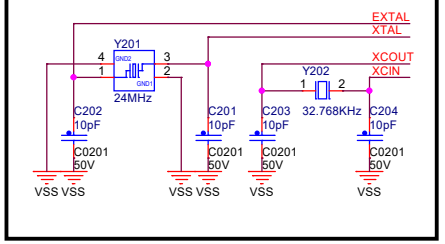
接口信号



通孔拓展



晶振

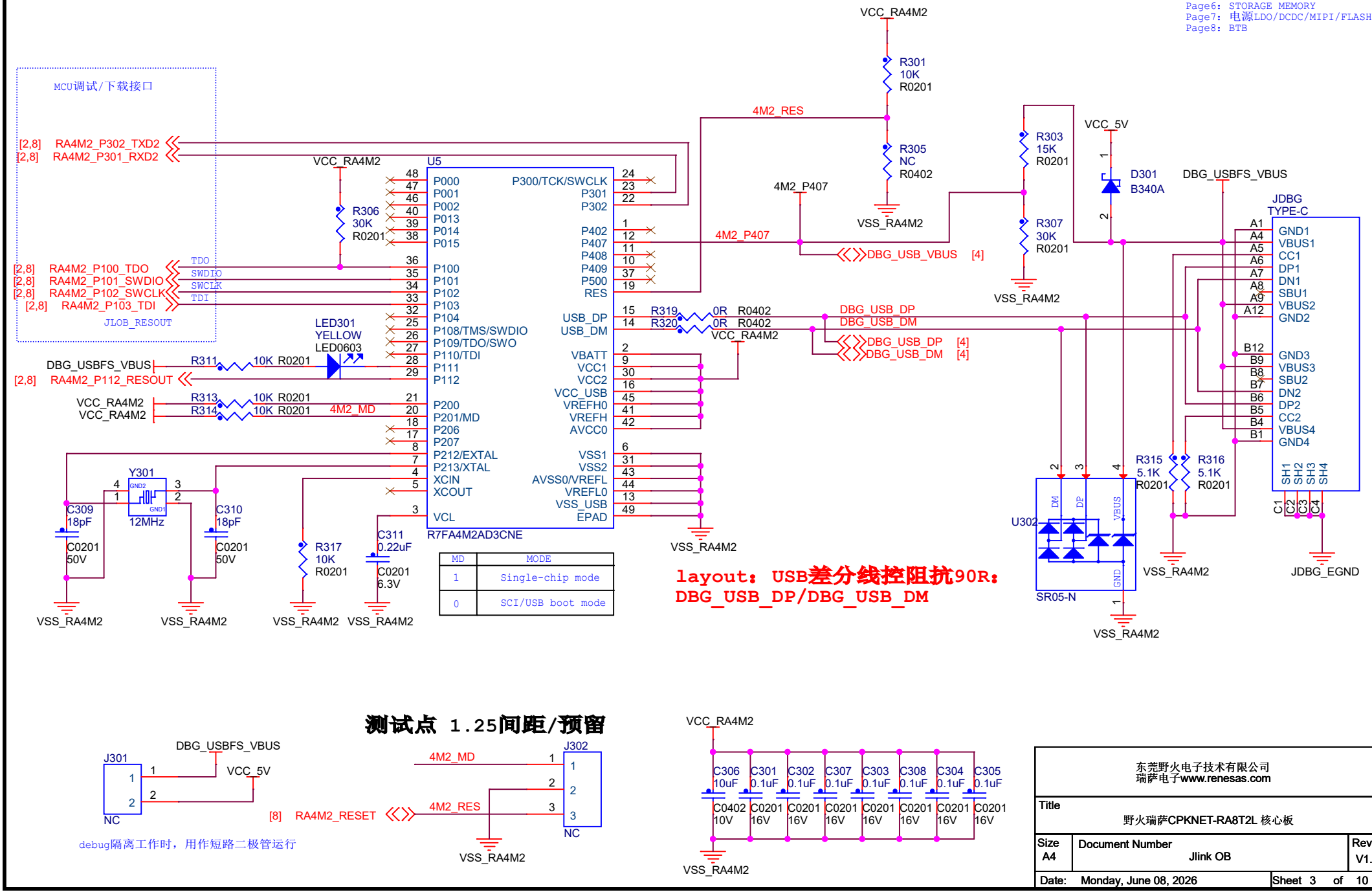


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Title: 野火瑞萨CPKNET-RA8T2L 核心板		
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Jlink OB

注意：R7FA4M2只能12MHz晶振

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Page8: BTB

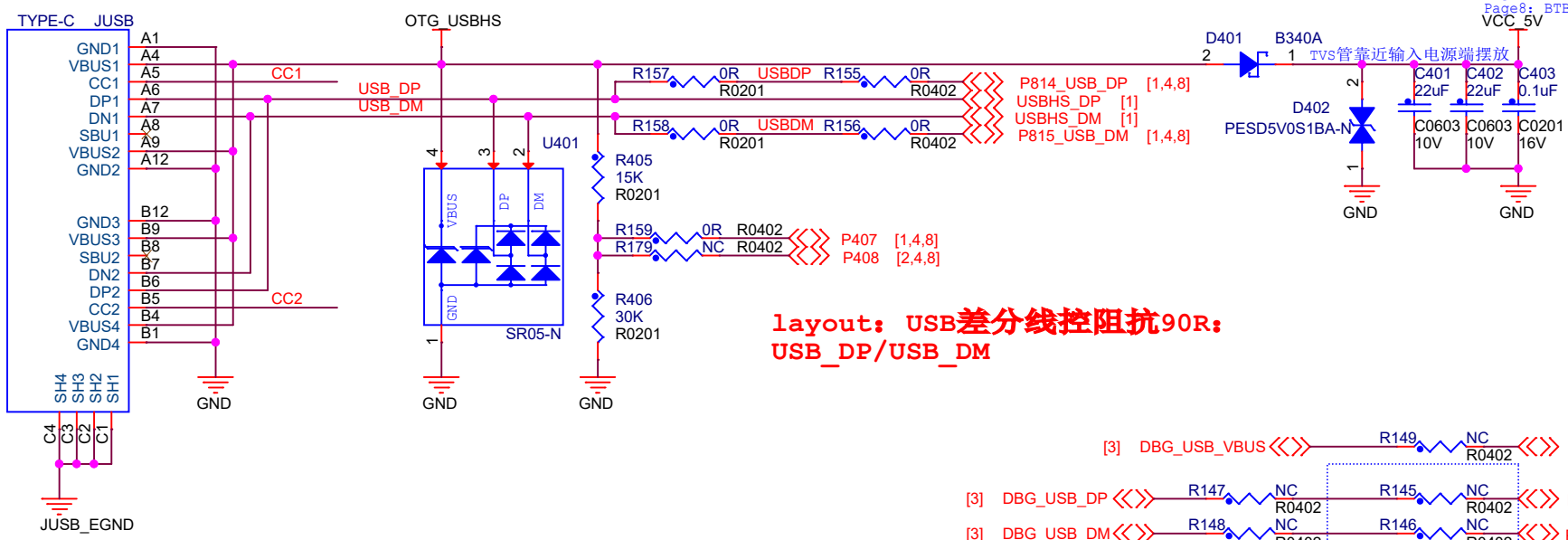


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Title	野火瑞萨CPKNET-RA8T2L 核心板
Size A4	Document Number Jlink OB
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Date: Monday, June 08, 2026	Sheet 3 of 10

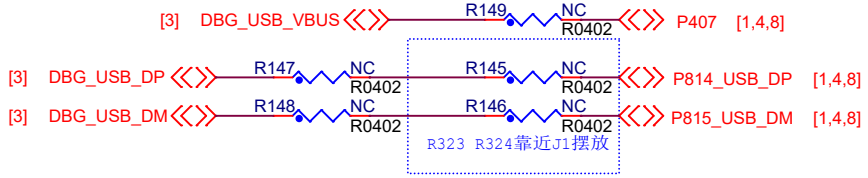
USB HS

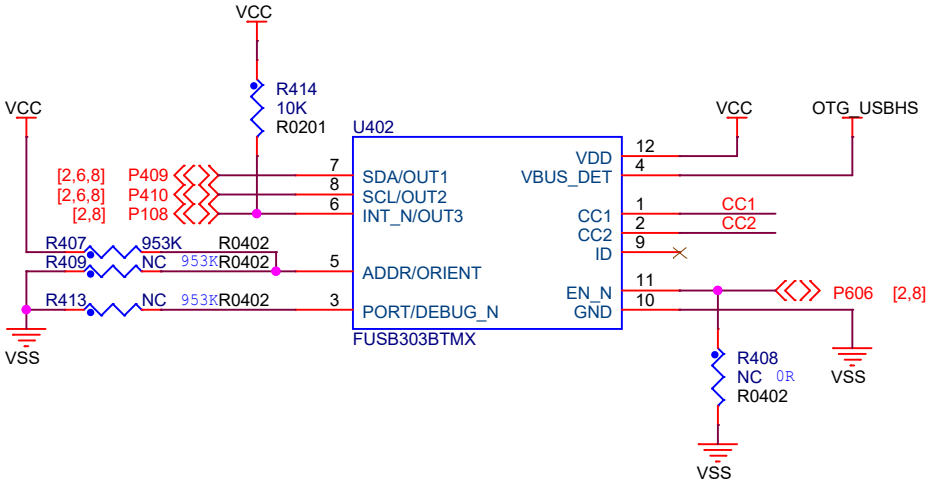
如果使用USB HS，主晶振只能选12/20/24/48MHz

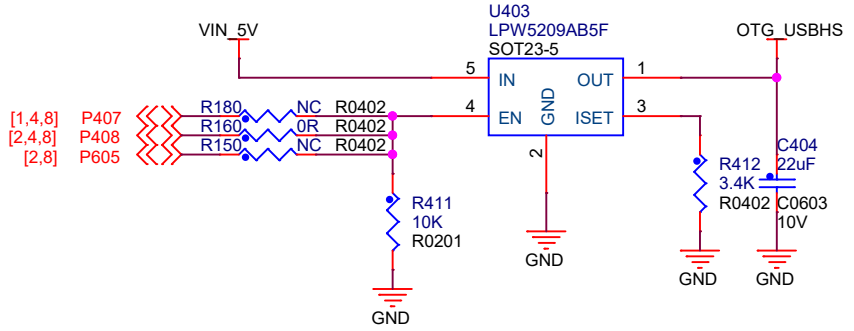
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layout：USB差分线控阻抗90R：
USB_DP/USB_DM





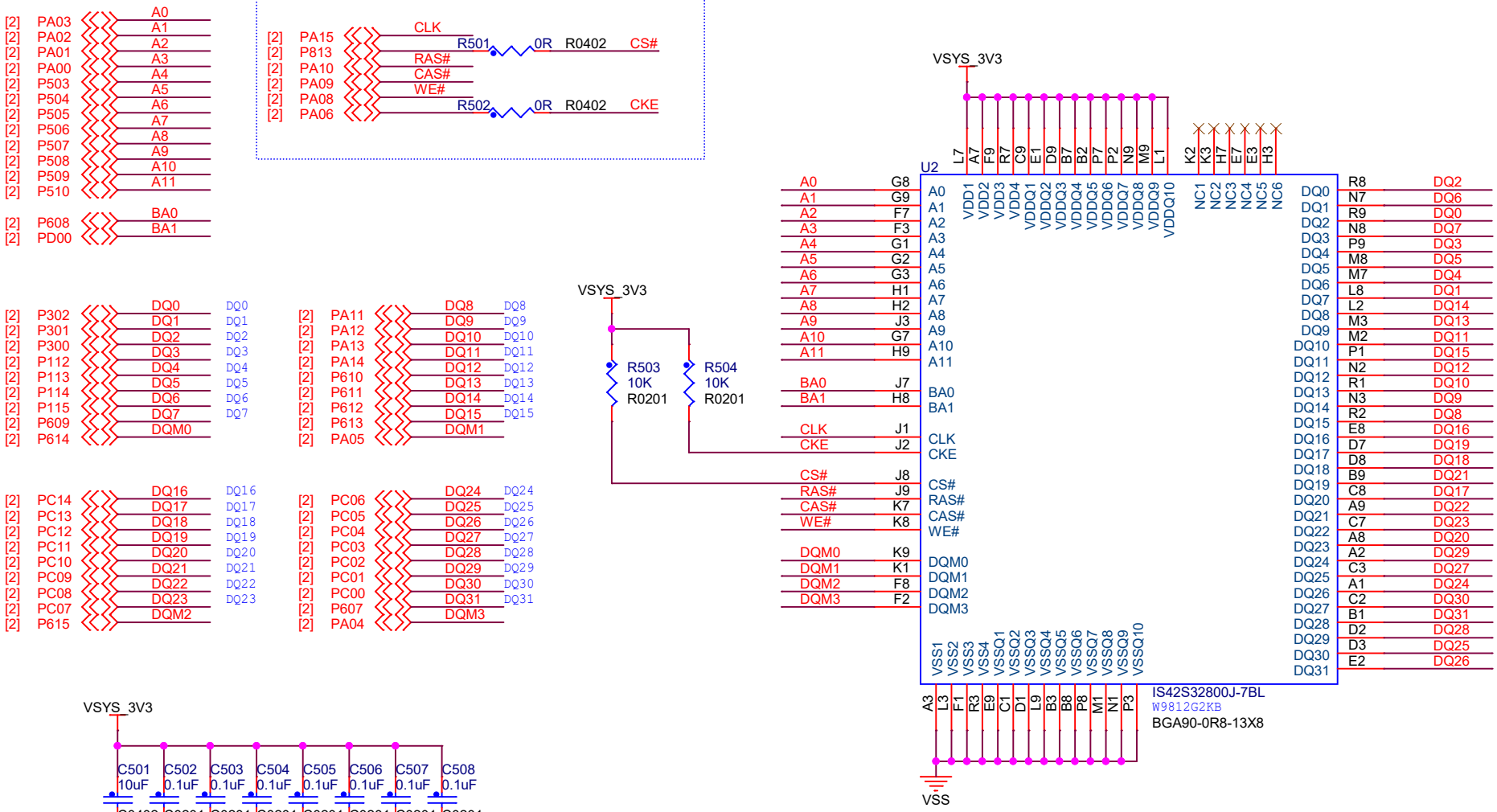


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Size: A4	Document Number: USB HS	Rev: V1.2
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SDRAM

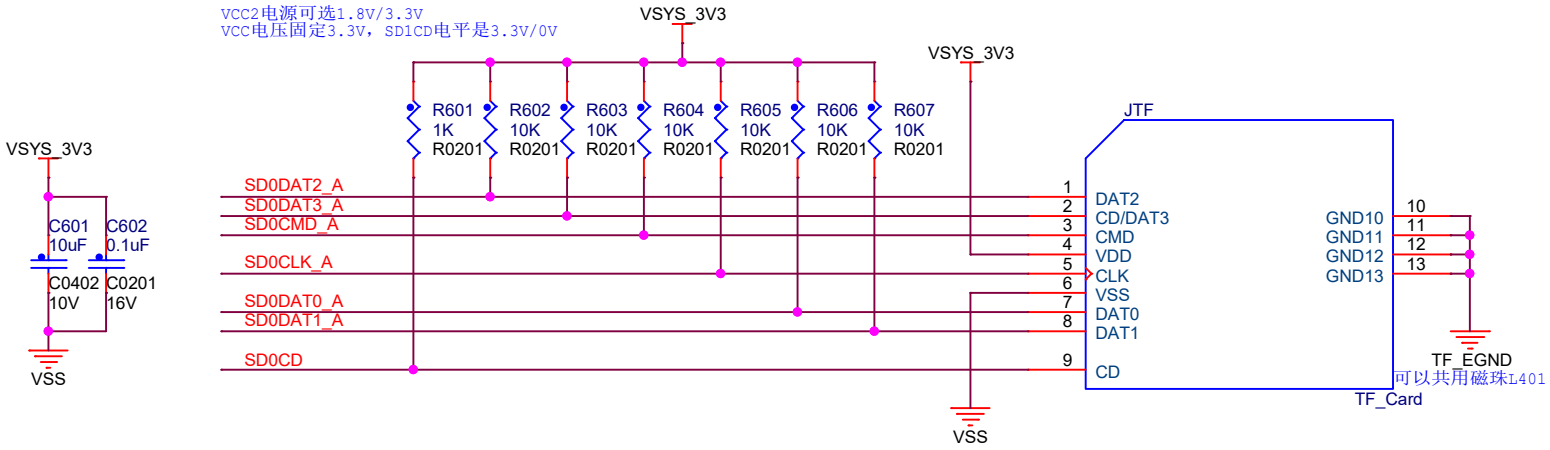
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layout： SDRAM走线控阻抗50R：



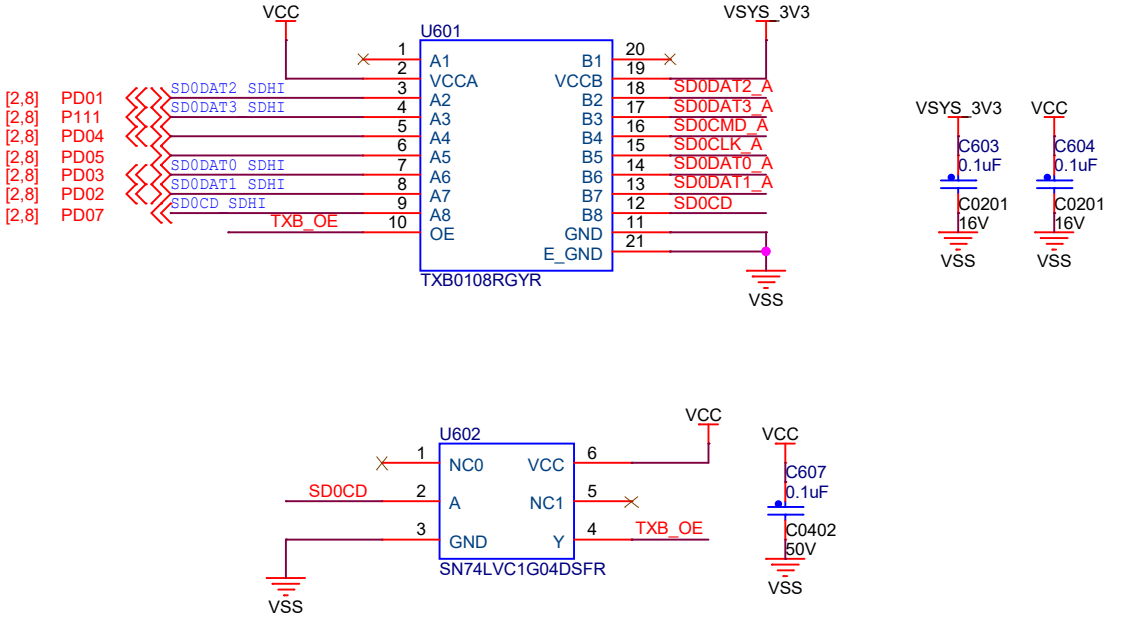
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Size: A4	Document Number: SDRAM	Rev: V1.1
Date: Wednesday, September 17, 2025	Sheet 5 of 10	

SD卡

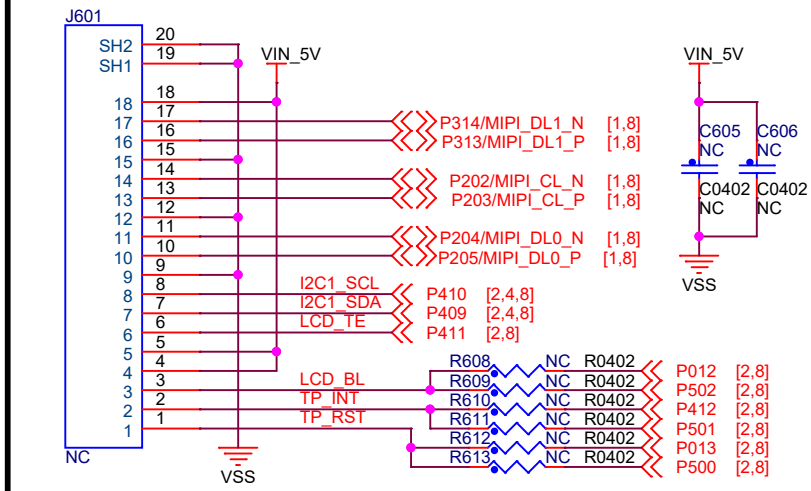


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SD卡电平转换

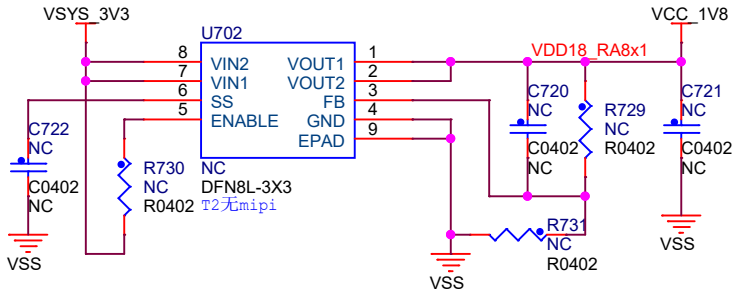


MIPI屏

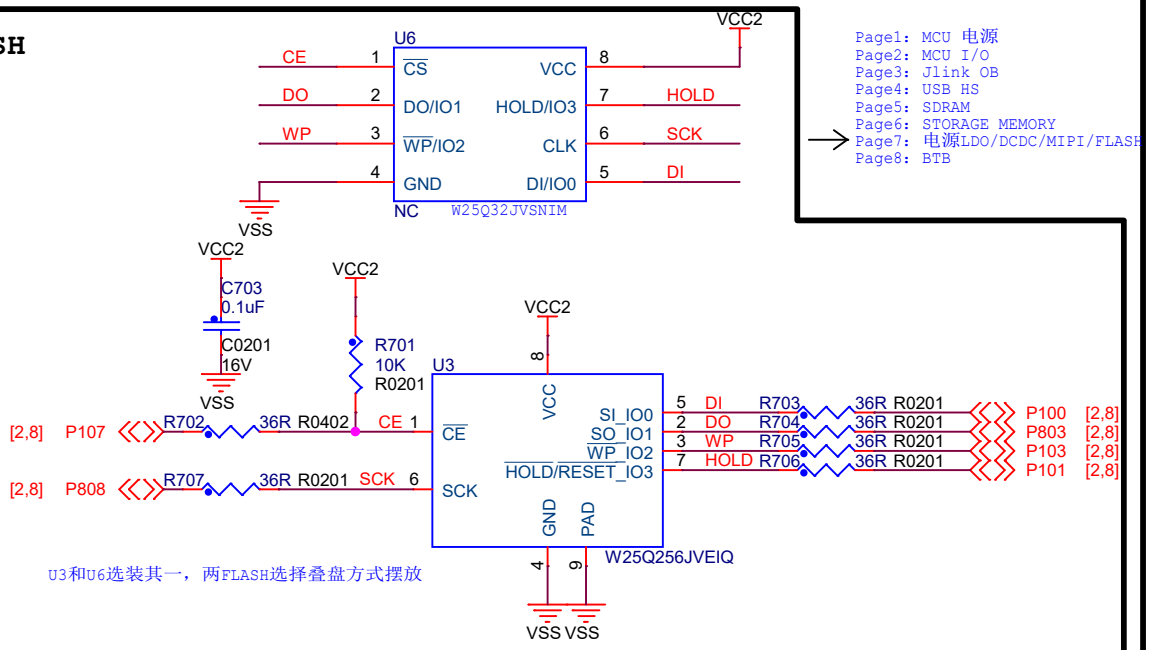


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Size: A4	Document Number: STORAGE MEMORY	Rev: V1.2
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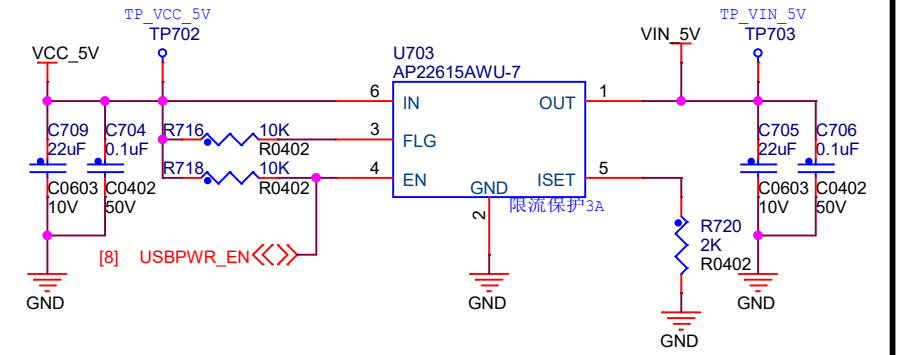
1.8V电源



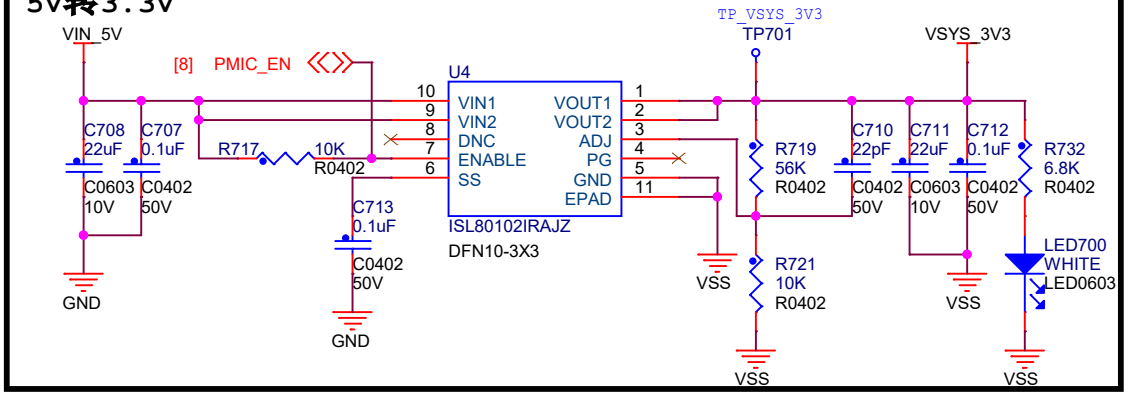
FLASH



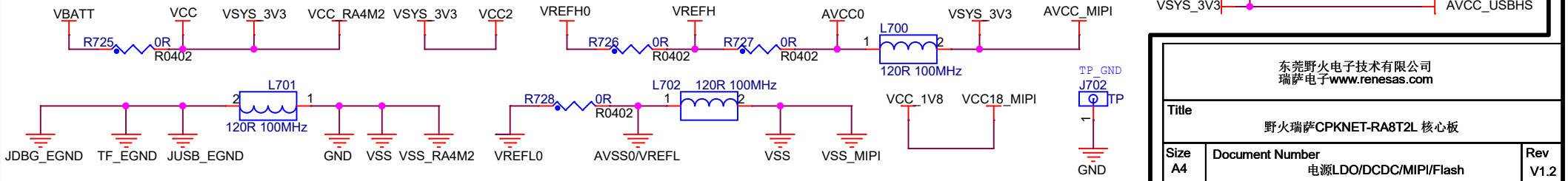
电源保护



5V转3.3V



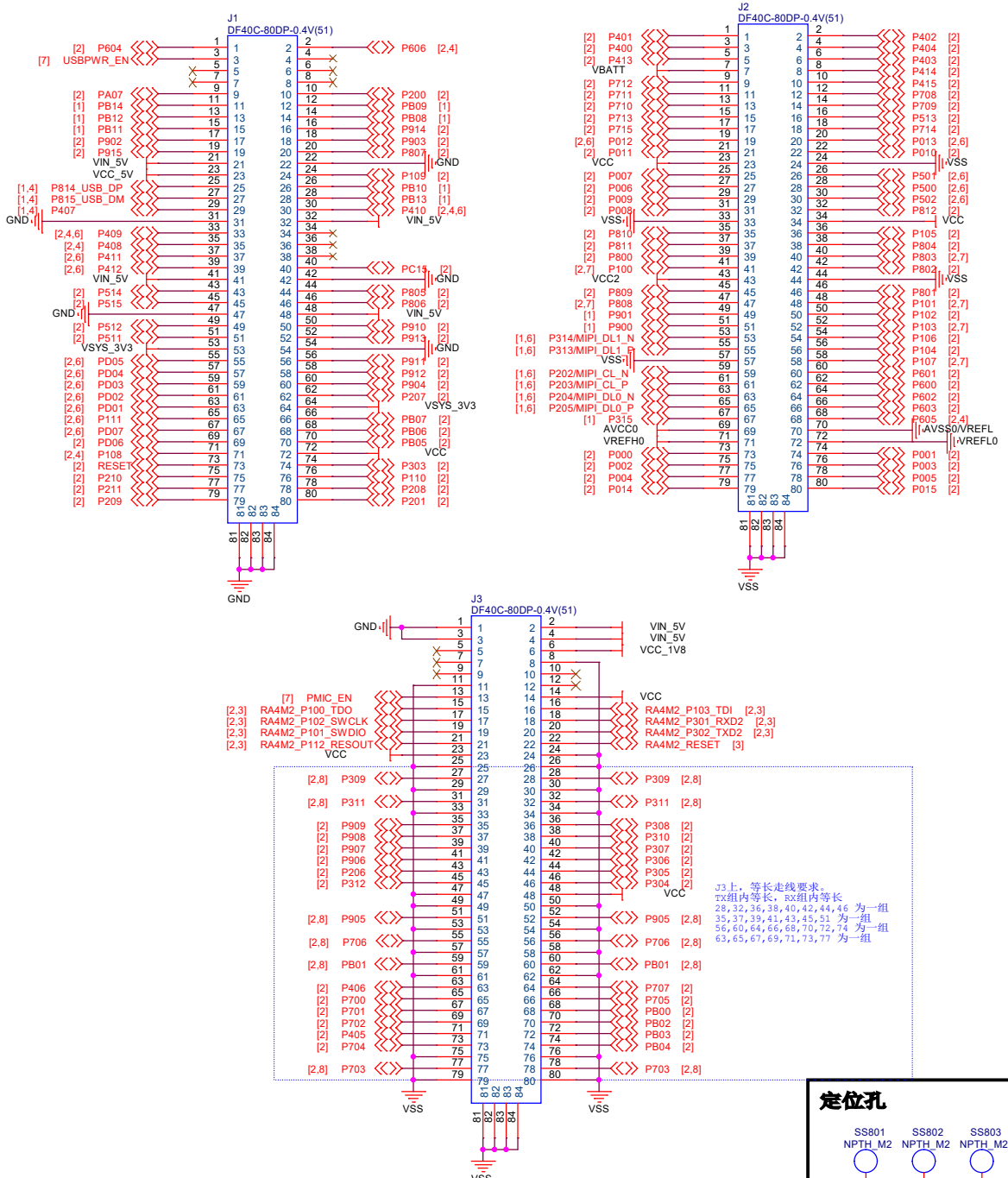
电源分割



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BTB座子

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Page8: BTB



定位孔

SS801	SS802	SS803	SS804
NPTH_M2	NPTH_M2	NPTH_M2	NPTH_M2

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历史版本

版本号	日期	设计	描述
V1.0	2025-08-28	LPC	初始版本
V1.1	2025-09-17	LPC	修正VCC_USB连接
V1.2	2026-06-08	llm	修改R205、R207和R601三个电阻阻值

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野火瑞萨CPKNET-RA8T2L 核心板_原理图_V1.1

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